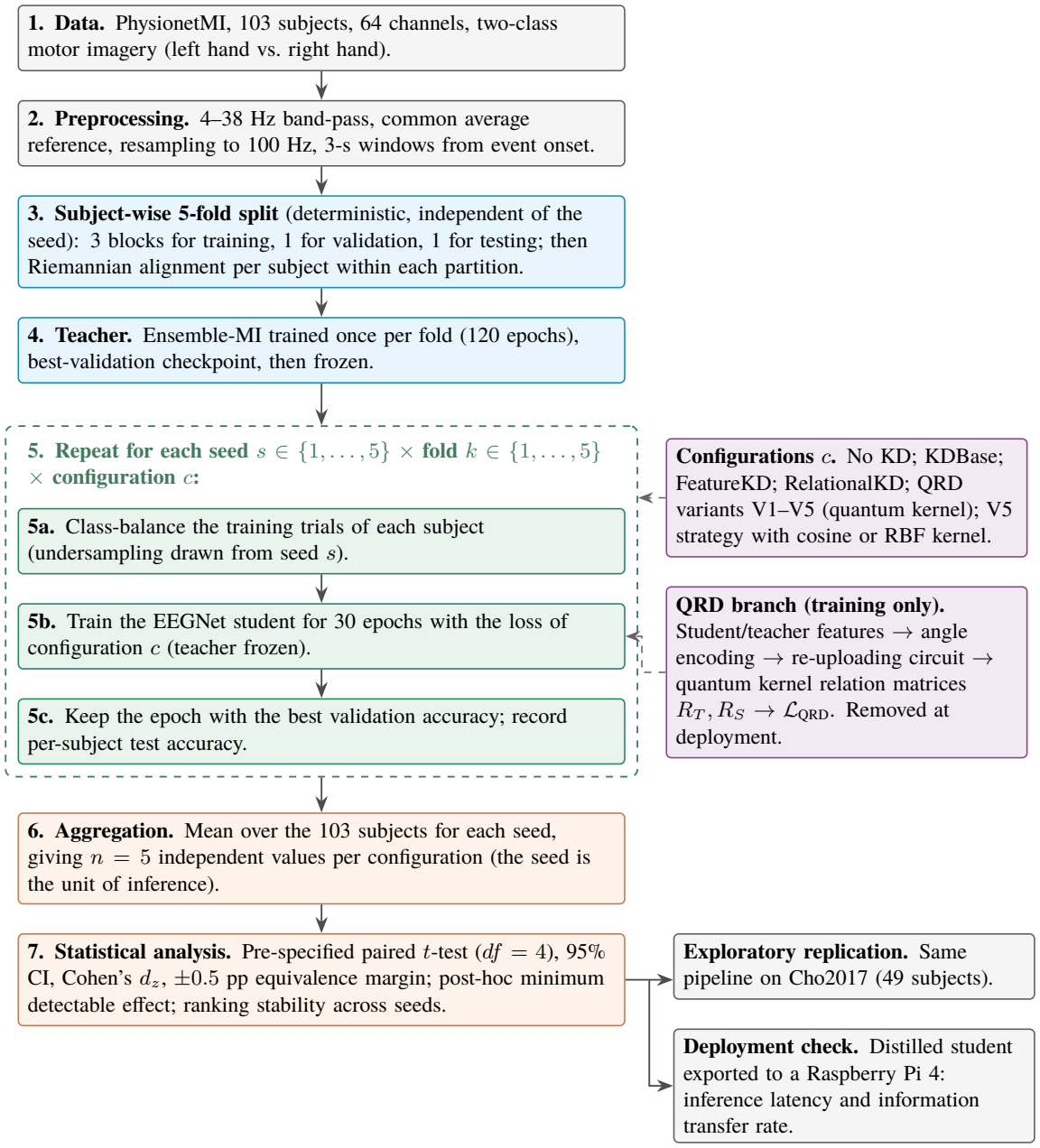
1. Data. PhysionetMI, 103 subjects, 64 channels, two-class motor imagery (left hand vs. right hand).
2. Preprocessing. 4–38 Hz band-pass, common average reference, resampling to 100 Hz, 3-s windows from event onset.
3. Subject-wise 5-fold split (deterministic, independent of the seed): 3 blocks for training, 1 for validation, 1 for testing; then Riemannian alignment per subject within each partition.
4. Teacher. Ensemble-MI trained once per fold (120 epochs), best-validation checkpoint, then frozen.
5. Repeat for each seed $s \in \{1, \ldots, 5\}$ × fold $k \in \{1, \ldots, 5\}$ × configuration c:
5a. Class-balance the training trials of each subject (undersampling drawn from seed s).
5b. Train the EEGNet student for 30 epochs with the loss of configuration c (teacher frozen).
5c. Keep the epoch with the best validation accuracy; record per-subject test accuracy.
Configurations c. No KD; KDBase; FeatureKD; RelationalKD; QRD variants V1–V5 (quantum kernel); V5 strategy with cosine or RBF kernel.
QRD branch (training only). Student/teacher features → angle encoding → re-uploading circuit → quantum kernel relation matrices $R_T, R_S \rightarrow \mathcal{L}_{\text{QRD}}$. Removed at deployment.
6. Aggregation. Mean over the 103 subjects for each seed, giving $n = 5$ independent values per configuration (the seed is the unit of inference).
7. Statistical analysis. Pre-specified paired t-test ($df = 4$), 95% CI, Cohen's d_z, ±0.5 pp equivalence margin; post-hoc minimum detectable effect; ranking stability across seeds.
Exploratory replication. Same pipeline on Cho2017 (49 subjects).
Deployment check. Distilled student exported to a Raspberry Pi 4: inference latency and information transfer rate.